Mars Rover Sample Collection Planning Using PDDL

CSL304

Artificial Intelligence Assignment

1 Comparison of Satisficing and Optimal Plans

The 'satisficing' problem uses the Lazy Greedy search algorithm with an FF heuristic to try and find a valid plan quickly. Whereas the 'optimal' problem uses an A* search with an LM-Cut heuristic to find the optimal plan with the minimal cost.

Below shows the results of the plan lengths found through both scenarios.

Table 1: Comparison of planning performance

Measurement	**Orig. Satisficing**	**Orig. Optimal**	**Ext. Satisficing**	**Ext. Optimal**
Plan length	33	25	46	40
Plan cost	33	25	46	40
States expanded	68	5,792	143	2,230,060
Search time (s)	0.002226	0.490556	0.003239	267.355778
Planner time (s)	0.28	0.67	0.17	267.57

The optimal plan consists of 25 actions where as the satisfactory plan consists of 33 actions. This is a reduction of approximately 24.24%.

The optimal plan consists of 40 actions where as the satisfactory plan consists of 46 actions. This is a reduction of approximately 13.04%.

The reason for this difference is because the satisfactory algorithm will always try to find a solution quickly, it doesn't look to reduce the amount of actions. This means it may take an alternative path which involves extra moves and deposits. The optimal configuration will look to search through the search space to find the most optimal plan with the least amount of actions. This can become more computationally expensive when the problem increases in size, especially in the extended scenario.

2 Use of Storage Bays and Drill in the Optimal Plans

There are two storage areas (Bay1 and Bay2) on the rover. Each can only hold one sample. Optimal plans will use both storage areas to carry more than one sample and return to the Base to deposit them.

2.1 Original Scenario

In the original scenario, the optimal plan uses the two bays in two collection groups:

- Bay1 stores Rock2 at SiteB, while Bay2 stores Rock4 at SiteD.
- On a later trip, Bay1 stores Rock5 at SiteF, while Bay2 stores Rock3 at SiteC.

This allows the rover to collect two samples before returning to the Base, reducing unnecessary return trips.

The drill is picked up at SiteE, after collecting Rock2 and Rock4. The rover then uses it to collect the hard samples, Rock5 and Rock3. The drill is therefore acquired only when it is needed for hard-sample collection.

2.2 Extended Scenario

In the extended scenario, the optimal plan uses both bays in three collection groups:

- Bay1 stores Rock4 at SiteD and Bay2 stores Rock2 at SiteB.
- Bay1 stores Rock5 at SiteF and Bay2 stores Rock7 at SiteF.
- Bay1 stores Rock3 at SiteC and Bay2 stores Rock6 at SiteE.

The rover returns to the Base and deposits the two stored samples after each collection group. This effectively uses both bays and avoids returning to the Base after collecting every individual sample.

The drill is picked up at SiteE after the rover has collected Rock4 and Rock2. It is then used to collect the hard samples: Rock5, Rock7, and Rock3. Rock6 is soft, so it can be collected without the drill.

3 Route Analysis in the Extended Scenario

In the extended scenario, the dust storm closes the route between Base and SiteB. Therefore, the rover cannot travel directly between them.

To reach SiteB, the optimal plan follows this route:

$$\text{Base} \rightarrow \text{SiteA} \rightarrow \text{SiteC} \rightarrow \text{SiteD} \rightarrow \text{SiteB}$$

The rover reaches SiteD through the one-way connection from SiteC to SiteD and then proceeds from SiteD to SiteB using the existing connection.

The one-way route is therefore used in the optimal plan. The connection allows travel from SiteC to SiteD, but not from SiteD back to SiteC. When returning from SiteB, the rover travels through SiteD to SiteE, then SiteC, SiteA, and finally Base. It does not attempt to use the prohibited reverse direction from SiteD to SiteC.

Thus, the optimal plan respects the changed connectivity after the dust storm while still allowing the rover to visit both SiteB and SiteD and complete the sample collection task.

4 Conclusion

This comparison shows that the optimal configuration achieves shorter plans but requires much more searching. Both the optimal plans use the two storage bays to collect the samples in pairs. They also go to SiteE to pick up the drill, before collecting the hard samples. The extended plan goes via SiteA, SiteC and SiteD to reach SiteB. This uses the one-way link between SiteC and SiteD correctly.